

A Neurosymbolic Approach for Landslide Risk Assessment

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Abstract

Landslide risk assessment remains a critical challenge, where current predictive models are still far from providing sufficiently reliable support for high-impact decisions affecting people, infrastructure, and territory. In this work, we propose a neurosymbolic architecture for landslide risk assessment that combines two neural pipelines: one for high-resolution spatial *susceptibility mapping* and one for ground deformation *forecasting* from InSAR time series, with a symbolic layer of strategic reasoning based on *Intuitionistic Alternating-time Temporal Logic* (IATL). IATL is well suited to this scenario because it models the progressive refinement of information and distinguishes between what is not yet proved and what is known to be false. Through an illustrative example, we show how the symbolic layer can extend the neural component in two ways: by supporting decisions in poorly represented scenarios and by filtering physically implausible outputs. The result is a decision-support framework in which predictions are complemented by formally checkable strategic properties and inspectable reasoning traces.

Keywords

Neurosymbolic AI, Responsible AI, Strategic Reasoning, Landslide, Machine Learning

1. Introduction

Landslides are among the most common natural hazards in Italy and worldwide, with recurring impacts on people, infrastructure, and territorial heritage [1, 2]. For this reason, landslide risk assessment is not only a matter of predicting where instability may occur. It also requires deciding when the available evidence is strong enough to justify an alert, which actors should intervene, and whether the resulting decision can be explained and checked [3]. In recent years, machine learning has contributed substantially to landslide susceptibility mapping, building on a long tradition of inventories and statistically based models [4, 5], and has grown with deep learning architectures, from convolutional networks to graph-based ones, up to transformer-like models on tabular data [6]. On the temporal side, satellite InSAR monitoring, based on techniques such as Permanent Scatterers and Small Baseline approaches [7, 8], has become central for investigating landslides and unstable slopes [9]; more recently, recurrent models, CNN-BiLSTM hybrids with attention, and explainable AI techniques have been explored for deformation forecasting from InSAR time series [10, 11, 12, 13]. These approaches, however, exhibit structural limitations when moving from prediction to decision. Modern neural models may be poorly calibrated, so that a high confidence score does not necessarily correspond to a reliable probability of correctness [14]; landslide datasets are frequently affected by strong imbalance between stable and unstable samples, distorting the behaviour of deep classifiers if not handled carefully [15]; and spatial and temporal dependence can make standard random validation overly optimistic, since performance measured on nearby pixels or adjacent time windows does not imply generalization to a new slope or a future deformation regime [16]. Most importantly, purely predictive models do not encode physical and contextual constraints: a susceptibility map can flag a pixel as high-risk in a flat area, or miss a critical situation simply because the site was not represented in the training set. In high-impact domains such as landslide risk management, *Responsible AI* requires decisions that are not only accurate but also verifiable, transparent, and robust to spurious outputs [17]. This is particularly demanding in territorial risk scenarios, where evidence arrives progressively: a new InSAR acquisition, a rainfall episode, or a

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field inspection can refine the current picture without making it complete. In such a setting, the absence of proof of instability should not be confused with proof of stability, and the decision-support system should make this distinction explicit. Here, formal methods for strategic reasoning play a central role. Originating in the verification of reactive systems through temporal logics [18], they have established themselves as a natural tool for modelling the interaction among agents that cooperate or compete under explicit constraints. Logics like ATL [19] and its extensions [20, 21, 22, 23, 24, 25, 26, 27, 28, 29] allow expressing coalition properties of the form “there exists a strategy for the agents that guarantees φ ”, and verifying them algorithmically on the system structure. Transferred to landslide risk, this viewpoint provides an adequate language to express questions of operational interest and to verify them in a certified way on top of neural evidence: is there a monitoring-and-intervention strategy that prevents the loss of a critical infrastructure? Can an alert be justified not only by a predicted score, but also by the strategic structure of the situation? This work fits within this perspective. We adopt a *decoupled* neurosymbolic architecture [30, 31], in which a neural layer produces predictive evidence that feeds a symbolic layer of strategic reasoning over a labelled graph; a scheme akin to the one recently proposed for explainable diagnosis based on Obstruction Logic [32, 33]. With respect to that line of work, we make two different choices: (i) the application domain is landslide risk monitoring, and (ii) the reasoning formalism is Intuitionistic ATL (IATL) [34] instead of Obstruction Logic. IATL incorporates intuitionistic logic to model the progressive refinement of information, with PTIME-complete model checking, and offers a native treatment of the epistemic dynamics of monitoring, in which a fact can still be *not proved* without being declared false. To the best of our knowledge, this is the first work to assess its applicative use in a territorial risk domain. We present a neurosymbolic architecture for landslide risk assessment (Sec. 2), an operational neural component producing spatial and temporal evidence, an IATL formalization of the decision problem (Sec. 3), and an example showing where neural reasoning stops and where the symbolic layer extends the system. We conclude by discussing the implications from a Responsible AI perspective (Sec. 4).

2. System Architecture and Neural Component

The proposed architecture follows the *decoupled* scheme typical of neurosymbolic systems: the neural and symbolic layers communicate through an interface of discrete evidence, but are designed and trained independently, so that neural models can be updated or replaced without modifying the symbolic theory, and vice versa. The core of the contribution lies in the symbolic layer (Sec. 3); here we describe the two neural pipelines that feed it and the evidence interface connecting them. For concreteness, we describe a typical use case, generalized from data available in scenarios of this kind. The **spatial pipeline** operates on a high-resolution raster dataset (metric grid derived from a DEM) over a region of interest at territorial scale: the dependent variable is the presence or absence of a landslide per pixel, derived from an inventory of mapped polygons (a few million usable pixels, with positive-class prevalence of about 8%); the independent variables are geomorphological descriptors derived from the DEM (slope, curvature, aspect encoding, and others). For each pixel, it returns a susceptibility classification on an ordinal scale (from very low to very high). The **temporal pipeline** uses time series from the European Ground Motion Service (EGMS) [35], a Copernicus product based on interferometric processing of Sentinel-1 acquisitions at a 12-day cadence, and produces for each InSAR measurement point a cumulative deformation forecast at 12 days. For both pipelines, we assume *spatially independent* and *temporally disjoint* splits to prevent any form of *leakage*. For the symbolic layer, two families of outputs matter, defined on each pixel (x, y) of the raster grid and, for the temporal component, on the time instant t : a **susceptibility class** $c_s(x, y)$ and an **expected deformation** $\widehat{d}(x, y, t)$ at 12 days. An **evidence extractor** transforms these into atomic propositions on a state graph, applying operational thresholds and integrating them with geomorphological priors and exposure information (critical infrastructure, built-up areas, road network). On this labelled graph operates an **IATL model checker** (Sec. 3), which verifies strategic properties of interest and, whenever a satisfying strategy exists, provides the operator with a witness.

3. Symbolic Component: IATL Strategic Reasoning

This section is the core of the contribution. We model the decision problem as a multi-agent system over a Birelational Concurrent Game Structure (BCGS), the native semantic structure of IATL; due to space constraints, we refer the reader to [36, 34] for the full syntax and semantics, and recall only the elements relevant to our modelling. Landslide risk monitoring is an intrinsically *epistemic* process: the object of reasoning is a slope with an evolving state, and the operator’s decisions (warning, evacuation, mitigation) face a non-controllable adversary, namely nature, under conditions of **partial information that is refined over time**. This is precisely the scenario for which IATL was introduced: its partial order $\preceq$ over states captures the notion of “more informative state”, and the intuitionistic interpretation of $\rightarrow$ guarantees monotonicity of the truth of formulas along the refinement. A new InSAR acquisition, a heavy rainfall, or an on-site inspection move the system from a state s to a state s' with $s \preceq s'$, in which propositions remain true but new ones can be proved. The fact that in IATL the law of excluded middle does not hold in general has a natural reading here: a state in which the slope is not proved to be unstable *is not equivalent* to a state in which it is proved to be stable, a crucial distinction in safety-critical contexts.

We define four agents: Nat (Nature: rainfall, kinematic evolution, seismic events), Mon (the monitoring system), Op (the decision-making operator, e.g. civil protection), and Civ (the civilians present on the exposed territory). The actions of Mon produce new evidence; those of Op issue alerts and order mitigation interventions; those of Civ express exposure behaviours (staying, moving away, evacuating); those of Nat represent the non-controllable evolutions of the environment. The explicit presence of Civ has a twofold motivation: civilians are the recipients of Op’s decisions, and therefore make properties of effectiveness verifiable, and they allow the formulation of **genuine coalition** properties $\{\text{Mon}, \text{Op}, \text{Civ}\}$ against Nat, in which safety emerges from the alignment of three actors.

A **state** s describes a territorial unit (a *slope unit* or a group of pixels) and includes the atomic propositions derived from neural evidence and priors. From the spatial pipeline: `high_susc`, `medium_susc`, `low_susc`. From the temporal pipeline: `accelerating`, `stable`, `gap`. From geomorphological priors: `flat_terrain`, `steep_slope`, `lithology_critical`. From territorial context: `urbanized`, `critical_infrastructure_downslope`, `evacuation_path_available`. From Op’s actions: `alert_issued`, `mitigation_active`. From Civ’s state: `civ_exposed`, `civ_evacuated`, `alert_received`. The **transitions** δ model the joint temporal evolution of observations, decisions, civilian responses, and environmental dynamics; the **partial order** $\preceq$ is the informational refinement ($s \preceq s'$ when s' contains all propositions proved in s plus possibly new ones). The monotonicity condition and the *well-behaved* conditions [34] hold by construction. The natural coalitions in this model are $\{\text{Op}\}$, $\{\text{Mon}, \text{Op}\}$, and $\{\text{Mon}, \text{Op}, \text{Civ}\}$, in opposition to Nat. We report three representative formulas.

φ_1 : **Effectiveness of the alert protocol**. An alert is effective only if it translates into safety behaviour. We express this property as a coalition guarantee:

$$\varphi_1 := \text{civ_exposed} \wedge \text{accelerating} \rightarrow \langle\langle \text{Mon}, \text{Op}, \text{Civ} \rangle\rangle \Diamond \text{civ_evacuated}. \quad (1)$$

The cooperation of all three is necessary: Op issues the alert, Mon transmits reliable information, and Civ responds. Smaller sub-coalitions do not suffice, and this is exactly what the model makes verifiable.

φ_2 : **Physical plausibility**. A qualitative constraint on the adversary, independent of any action of the coalition:

$$\varphi_2 := \text{flat_terrain} \rightarrow \neg \langle\langle \text{Nat} \rangle\rangle \Diamond \text{landslide_event}. \quad (2)$$

On flat terrain, Nature has no strategy leading to a landslide. The constraint is qualitative, not probabilistic, and once verified in s it remains valid throughout the *cone* of refinements of s .

φ_3 : **Accountability in compound scenarios**. The joint duty of monitoring and operator in the presence of downslope exposure:

$$\varphi_3 := \langle\langle \text{Mon}, \text{Op} \rangle\rangle \Box (\text{civ_exposed} \wedge \text{accelerating} \rightarrow \text{alert_issued} \wedge \text{alert_received}) \quad (3)$$

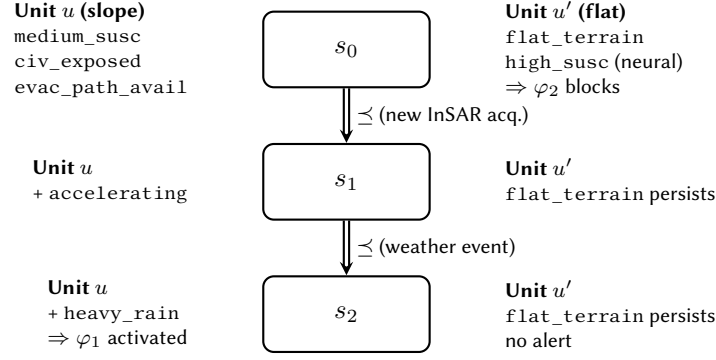


Figure 1: Minimal BCGS for the example. On unit u (left), the accumulation of evidence activates φ_1 ; on unit u' (right), the constraint φ_2 blocks the spurious alert.

The difference with φ_1 is crucial: φ_3 is a *process* property (the alert must be issued and received), φ_1 is an *outcome* property (evacuation must take place). Together they capture two complementary levels of accountability. The complexity of IATL model checking is PTIME-complete [34], so verification remains practicable even on realistic BCGSs. Consider (Fig. 1) a portion of territory with two adjacent units: u , a sloping hillside with civilians downstream, and u' , a neighbouring flat area served by the same pipeline. The chain $s_0 \preceq s_1 \preceq s_2$ describes the informational evolution: in s_0 the propositions $\text{medium_susc}(u)$, $\text{civ_exposed}(u)$, $\text{evac_path_avail}(u)$ are known, and for u' both $\text{flat_terrain}(u')$ and the spurious indication $\text{high_susc}(u')$ hold; in s_1 a new acquisition reveals $\text{accelerating}(u)$; in s_2 a meteorological event adds heavy_rain across the whole area. **(a) Out-of-distribution generalization.** On u in s_2 , accelerating and heavy_rain co-occur in a combination poorly represented in the training data, on which the susceptibility estimate remains at $\text{medium_susc}(u)$: the neural outputs alone do not yield a univocal operational indication. The symbolic layer, by integrating a domain rule of the form $\text{accelerating} \wedge \text{heavy_rain} \rightarrow \text{prealert_required}$, enables the verification of φ_1 , and the coalition $\{\text{Mon}, \text{Op}, \text{Civ}\}$ has a strategy that certainly leads to $\text{civ_evacuated}(u)$. The *witness* is a sequence of joint actions directly inspectable by the operator. Generalization does not occur through interpolation in the data, but through **logical composition of evidence** over the refinement structure. **(b) Filtering of spurious outputs.** On u' , from s_0 onward, $\text{flat_terrain}(u')$ holds together with the neural signal $\text{high_susc}(u')$. The formula φ_2 comes into play as a qualitative constraint: the *cone* of refinements of u' contains no state reachable by Nature in which a landslide event occurs, and φ_3 is not activated on u' . The symbolic constraint **filters the spurious output** at the system level, without modifying the neural pipeline. In both cases the symbolic layer does not replace the neural one, but **selectively qualifies** its outputs: on u it amplifies a weak signal, on u' it disables a strong but implausible one. The final decision is always accompanied by the model checker’s trace, a concrete basis for the Responsible AI perspective discussed in the next section.

4. Responsible AI Perspective and Conclusions

We have presented a neurosymbolic architecture for landslide risk, in which two neural pipelines for *susceptibility mapping* and *deformation forecasting* feed a layer of strategic reasoning in Intuitionistic ATL, used to verify properties of operational interest on top of neural evidence. The choice of IATL is motivated by the epistemic nature of the domain: monitoring is a process of progressive evidence accumulation, and informational *refinement* is a first-class citizen of the model. The three typical requirements of Responsible AI [17] find concrete grounding here. **Verifiability:** properties such as φ_1 , φ_2 , and φ_3 are certified on the system structure by an IATL model checker in polynomial time [34], rather than declared a posteriori. **Transparency:** each decision is accompanied by a *witness* inspectable by the operator (Sec. 3), showing which neural evidence, symbolic constraints, and coalition actions led to the conclusion. **Robustness:** constraints such as φ_2 filter physically implausible neural signals

at the system level, without modifying the neural pipeline. Intuitionism also offers a more honest treatment of uncertainty, since absence of proof of instability does not amount to proof of stability: in safety-critical contexts, this distinction is a prerequisite for decision responsibility. As **future work**, we plan the *end-to-end* implementation and its evaluation on a real operational scenario; a further direction concerns a fuzzy extension of IATL, in which atomic propositions and transitions carry graded truth values [37, 38]. Building on fuzzy relational neural architectures [39], this would let the symbolic layer natively ingest the calibrated confidence of neural predictions and reason over degrees of evidence, tightening the integration between neural outputs and strategic reasoning.

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Declaration on Generative AI

The author(s) have not employed any Generative AI tools.

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